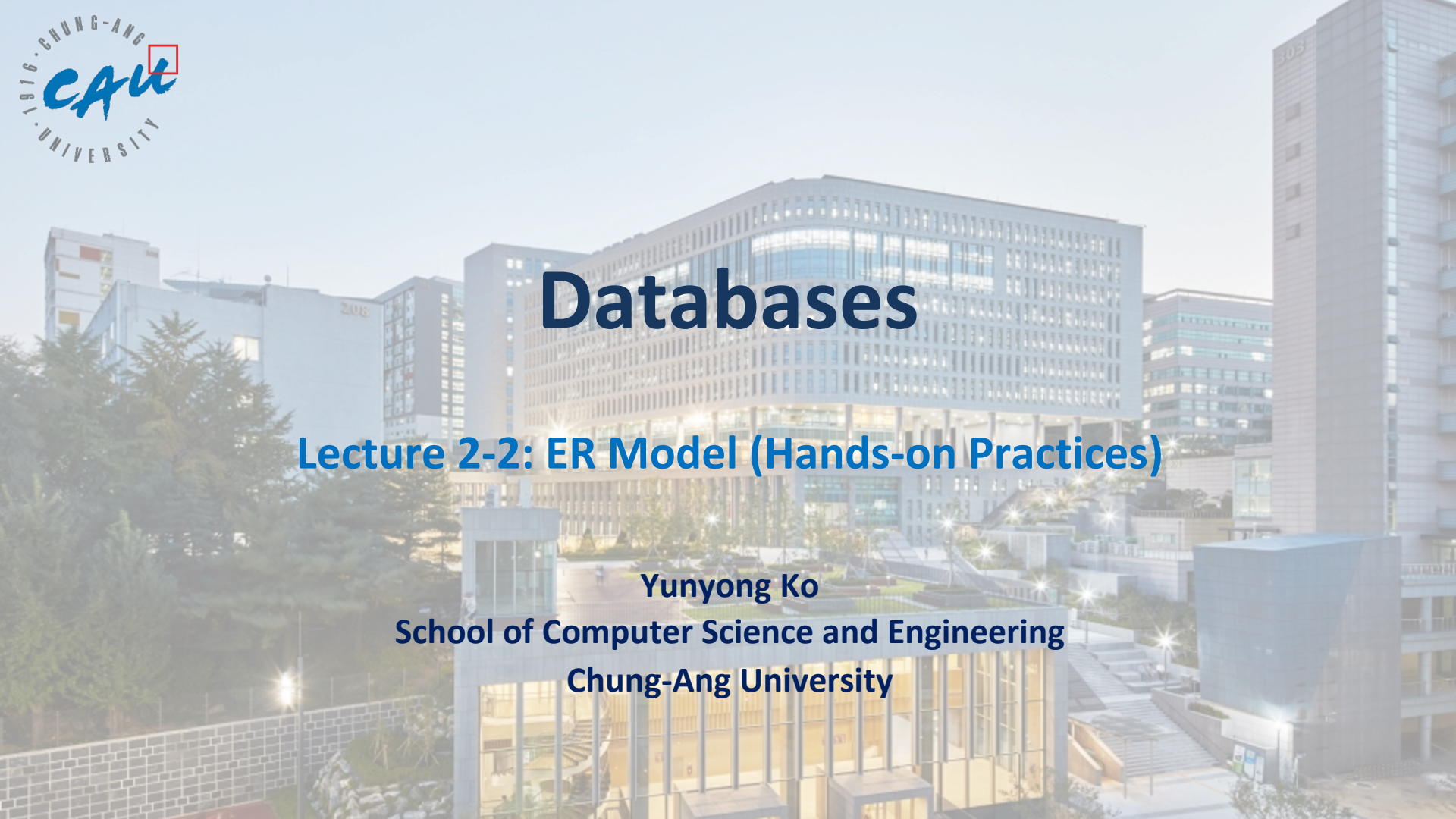


Databases

Lecture 2-2: ER Model (Hands-on Practices)

Yunyong Ko
School of Computer Science and Engineering
Chung-Ang University

Outline

- **Recap: Conceptual Data Modeling**
  - Entity-Relationship (ER) Model
  - Designing ER Schema



- **Practices: Designing ER Schema**
  - Draw.io installation
  - Exercise 1: MOVIE Database
  - Exercise 2: CONFERENCE_REVIEW Database

Recap: Entity-Relationship (ER) Model

❑ Describe Data as a Collection of *Entities* and *Relationships*

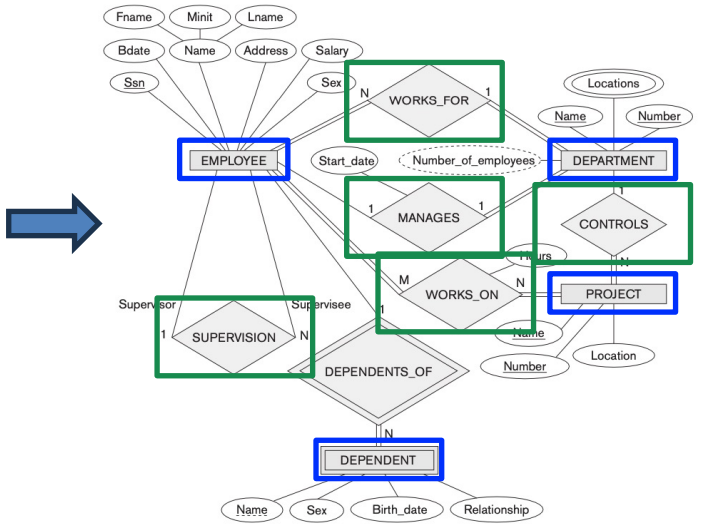
- **Entity**: a "thing" or "object" that is distinguishable from others
- **Relationship**: an association among several entities

❑ Focus on the Schema, Rather Than the Instances

❑ What Information Should be Managed?

- We want to keep track of a company's employees, departments, and projects
- **Department**
  - The company is organized into '**department**'s
    - Each has a unique name, a unique number, locations, manager, start date when that manager began, …
- **Project**
  - A department can control a number of '**project**'s
    - Each has a unique name, a unique number, a single location, controlling department
- **Employee**
  - '**Employee**'s are working in the company
    - Each has a name, SSN, address, salary, gender, birth date, projects that the employee works on, …
- **Dependent**
  - For insurance purposes, the '**dependent**'s of each employee are needed
    - Each has a first name, gender, birth date, relationships to the employee, …

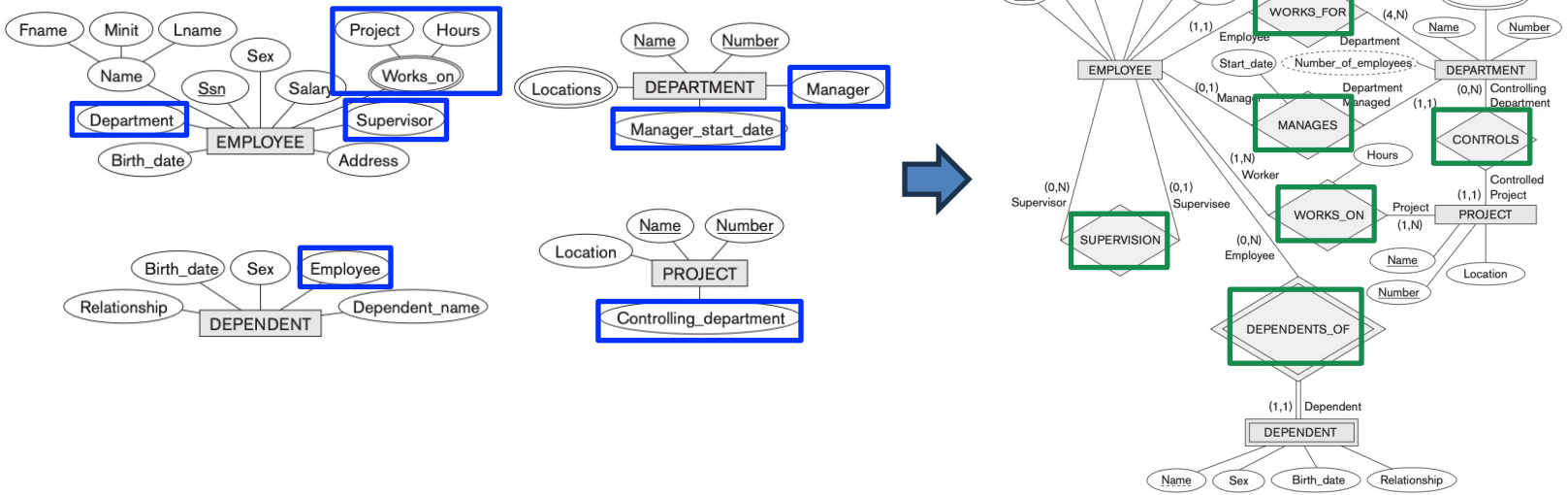
Requirement Analysis



Recap: Designing ER Schema

❑ Process of Designing ER Schema Diagram

- (1) Describe entity types and their attributes based on the data requirements
- (2) **Specify relationship types and their constraints**
- (3) **Change the attributes representing relationships into relationship types**



Recap: Constrains on Relationship Types

Constraints of Relationship Types

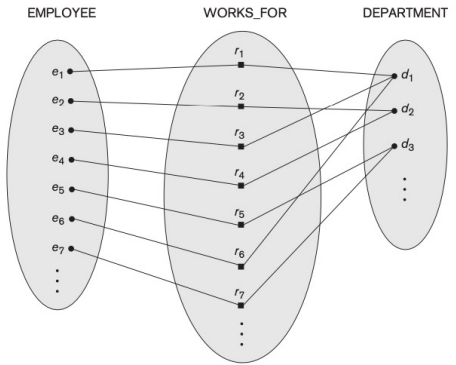
- Relationship types usually have certain constraints about participating entities
  - e.g., the company has a rule that each employee must work for exactly one department

Constraint 1: Cardinality Ratio

- Specify the maximum number of relationship instances that an entity can participate in
- Three types of cardinality ratios for a binary relationship
  - 1:1, 1:N, M:N

Constraint 2: Participation Constraint

- Specifies whether it should participate in a relationship type
- Two types of a participation constraint
  - **Total**: every entity should participate in the relationship
  - **Partial**: every entity does not need to participate in the relationship

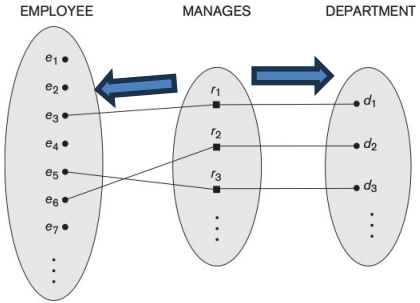


Recap: Attributes of Relationship Types

❑ Relationship Types Can Have Attributes

The company wants to keep track of:
(1) the **start date** of a manager when the employee manage the department;
(2) the **start date** of an employee when the employee works for the department
(3) the **number of working hours** of each employee on a specific project and

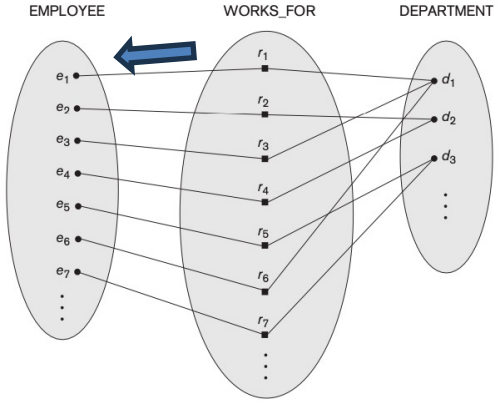
1:1 Relationship Type



StartDate for MANAGES

Migrated to **one of any participating entities**

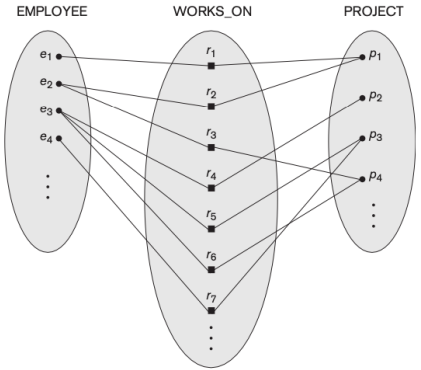
1:N Relationship Type



StartDate for WORKS_FOR

Migrated only to **the N-side participating entity**

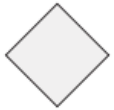
M:N Relationship Type

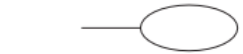


Hours for WORKS_ON

Cannot be migrated to any entities

Recap: Notations of ER Model

Symbol	Meaning
	Entity
	Weak Entity
	Relationship
	Indentifying Relationship



Attribute



Key Attribute



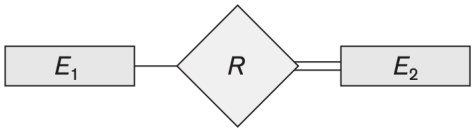
Multivalued Attribute



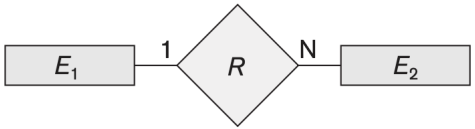
Composite Attribute



Derived Attribute



Total Participation of E_2 in R



Cardinality Ratio 1: N for $E_1 : E_2$ in R

Outline

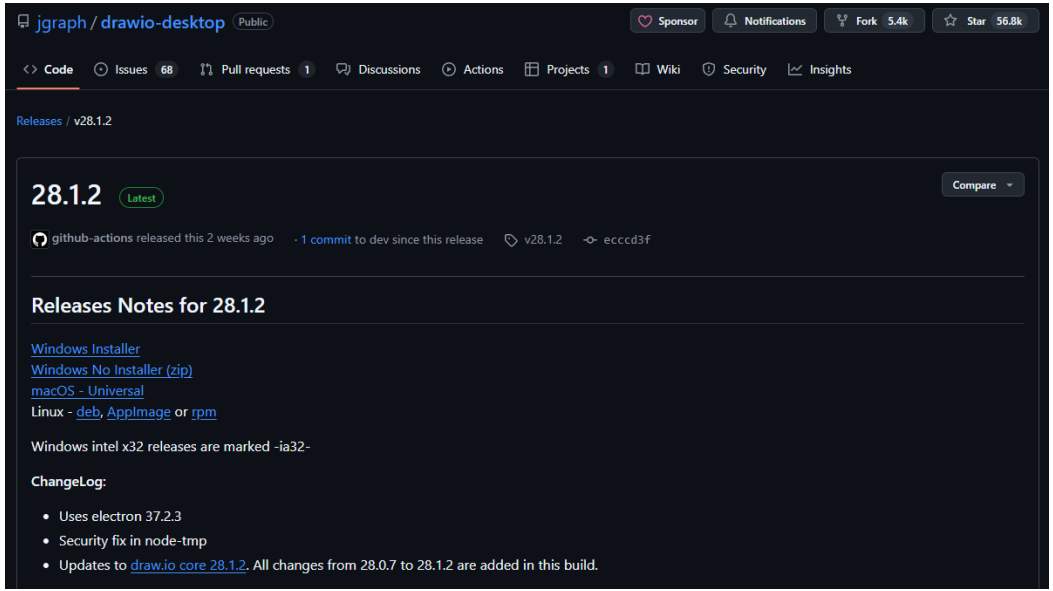
- Recap: Conceptual Data Modeling
  - Entity-Relationship (ER) Model
  - Designing ER Schema

- **Exercises: Designing ER Schema**
  - Draw.io installation
  - Exercise 1: MOVIE Database
  - Exercise 2: CONFERENCE_REVIEW Database



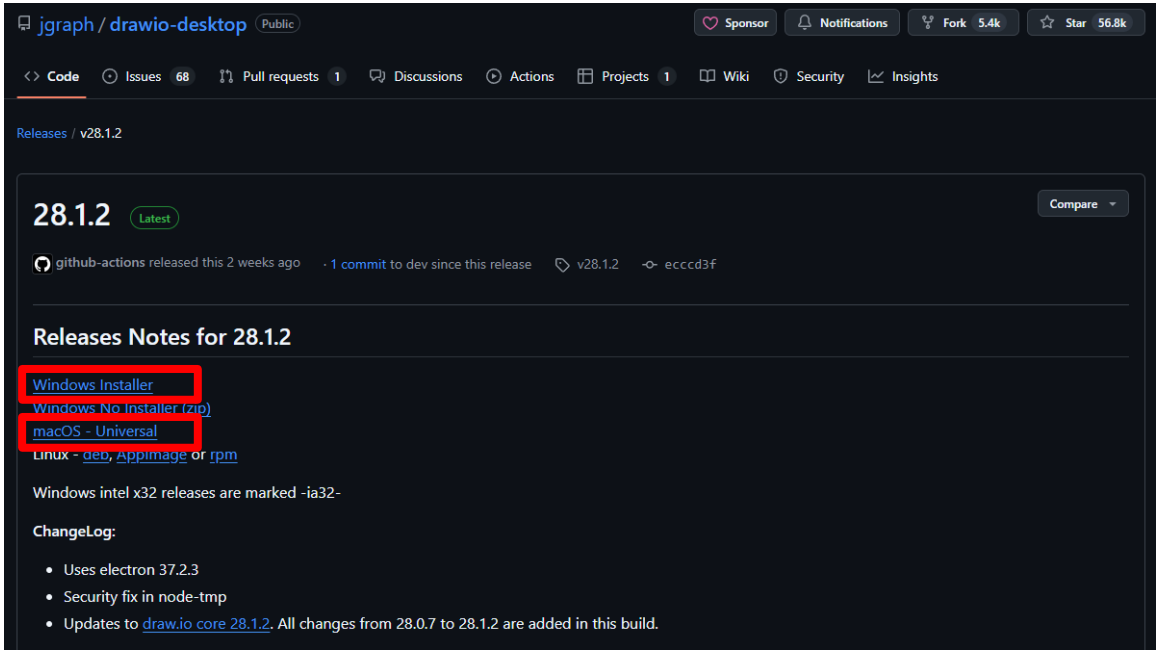
Draw.io Desktop Version Installation

- **Access the latest version from the draw.io desktop GitHub releases page**
  - https://github.com/jgraph/drawio-desktop/releases/tag/v28.1.2



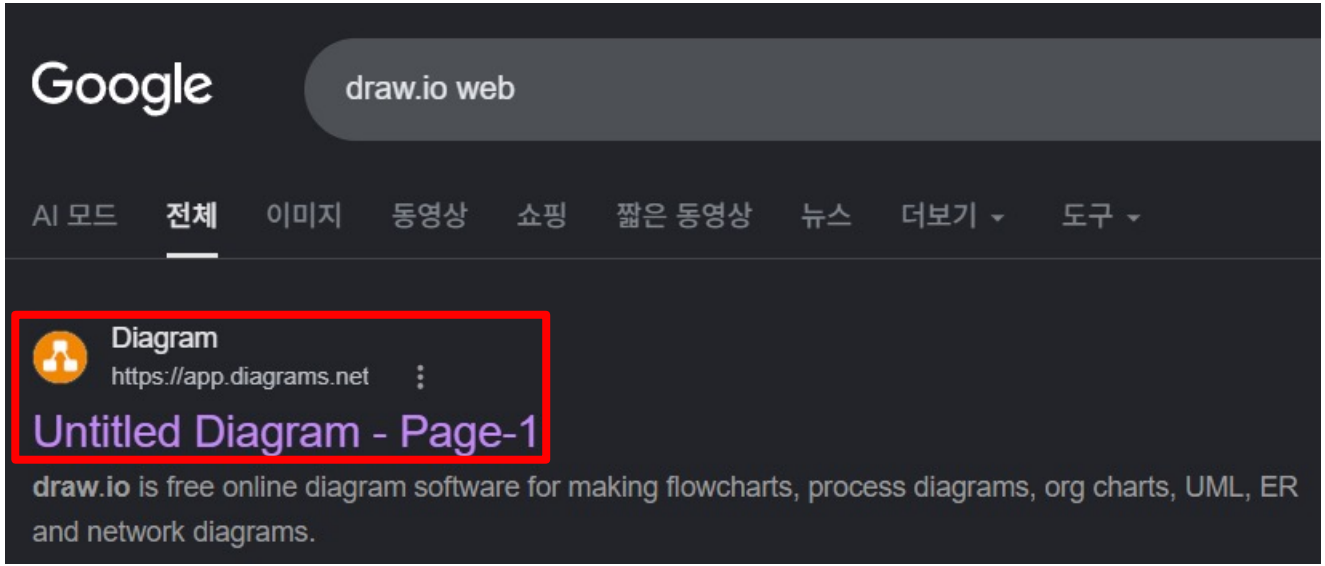
Draw.io Desktop Version Installation (Cont.)

- Click the appropriate link for your operating system and install



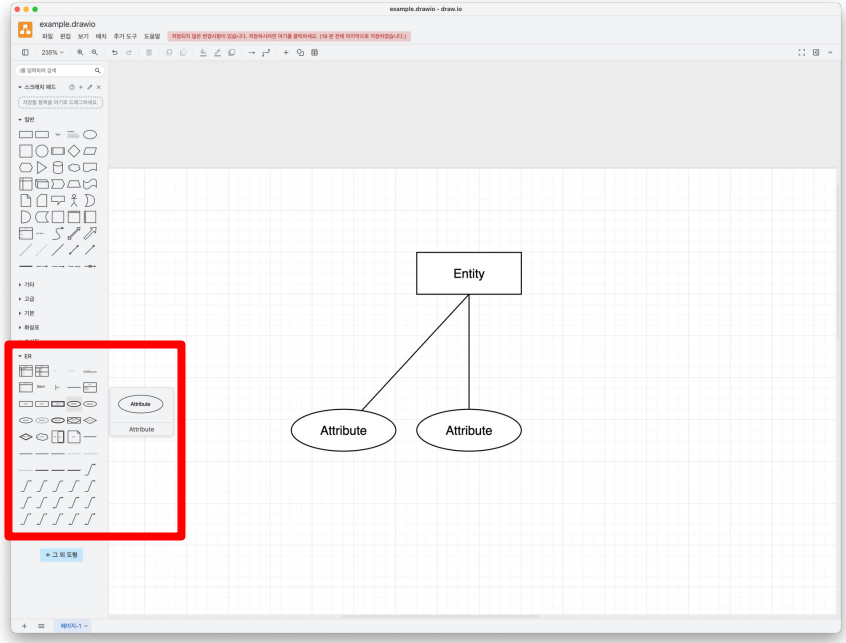
Draw.io Mobile (No Laptop)

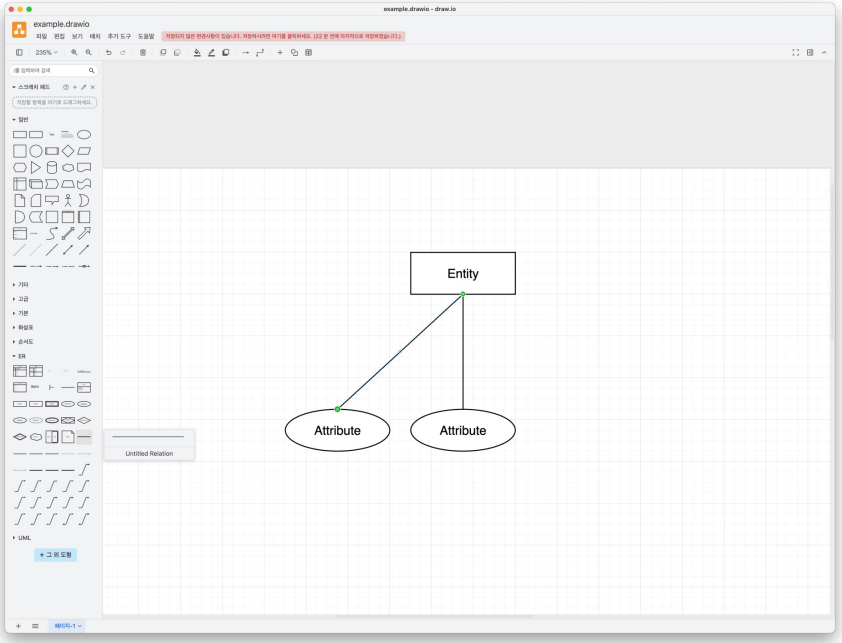
- Search for draw.io web on google and click the link



Draw.io Simple Tutorial

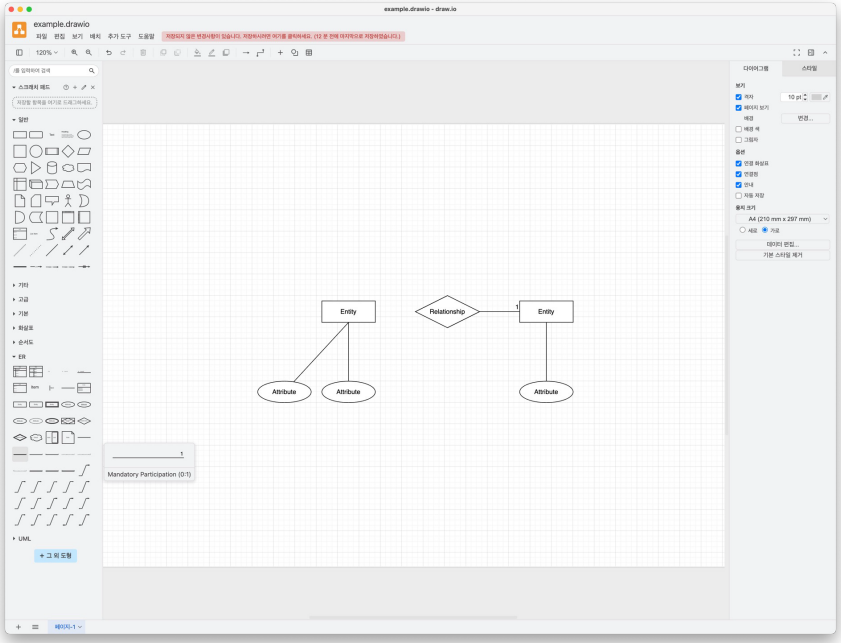
- Click the ER toggle in the left side panel to access predefined shapes such as Entity and Attribute

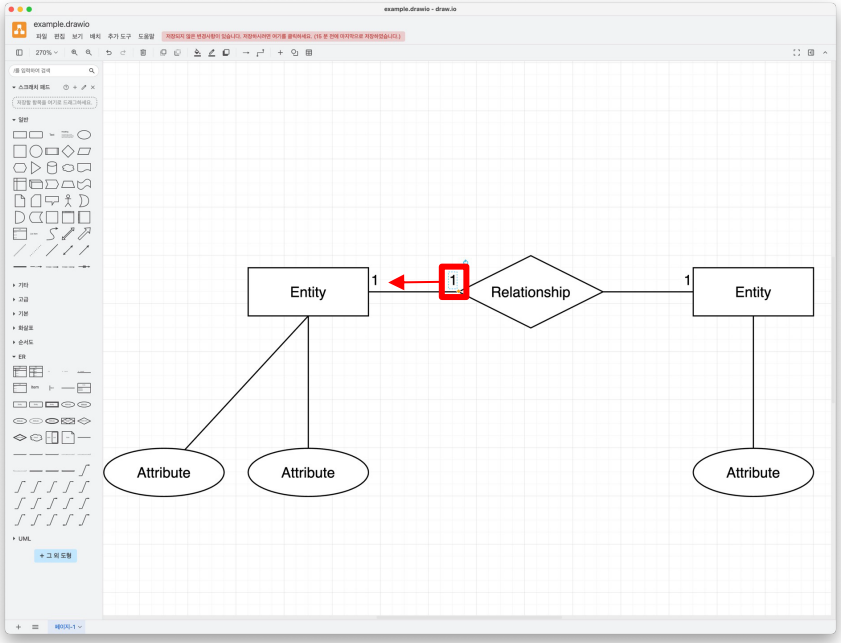




Draw.io Simple Tutorial

- You can connect relations with cardinality and adjust the position of the cardinality labels





Exercise 1: ER Schema Diagram

❑ Designing ER Diagram for MOVIE Database

- Consider a MOVIE database in which data is recorded about the movie industry.
  - ❑ MOVIE
    - Each movie is identified by (title and year of release)
    - Each has a length in minutes and a plot outline
    - Each is classified under one or more genres (e.g., horror, action, drama, …)
    - Each movie has a production company, one or more directors, and one or more actors appearing in it



  - ❑ ACTOR
    - Actors are identified by (name and date of birth) and appear in one or more movies
    - Each actor has a role in the movie and has zero or more quotable quotes to speak



  - ❑ DIRECTOR
    - Directors are identified by (name and date of birth) and direct one or more movies
    - It is possible for a director to act in a movie (including one that she/he may also direct)



  - ❑ PRODUCTION
    - Production companies are identified by (name) and each has an address
    - A production company produces one or more movies



Exercise 1: ER Schema Diagram (Cont.)

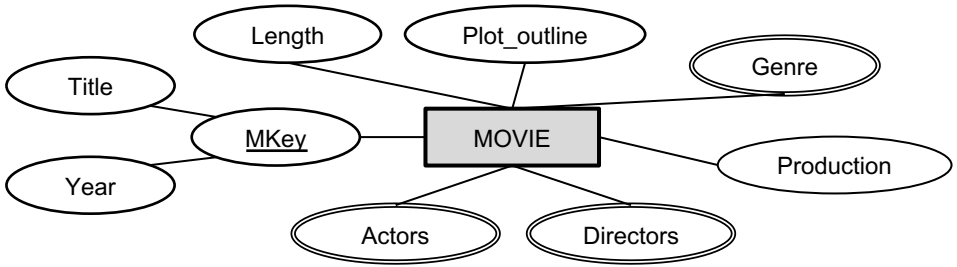
❑ Step 1: Describing the Entity Types

- List entity types and their attributes based on the data requirements

❑ Entity Type 1: MOVIE



- Requirements:
  - Each movie is identified by (title and year of release)
  - Each has a length in minutes and a plot outline
  - Each is classified under one or more genres (e.g., horror, action, drama, …)
  - Each movie has a production company, one or more directors, and one or more actors appearing in it
- Attributes:
  - *Mkey(Title, Year), Length, Plot_outline, {Genre}, Production, {Directors}, {Actors}*

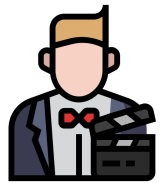


Exercise 1: ER Schema Diagram (Cont.)

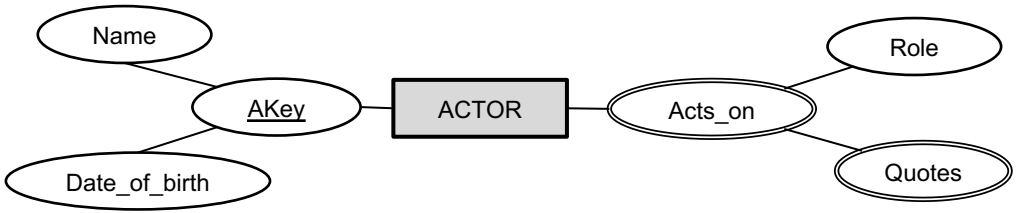
❑ Step 1: Describing the Entity Types

- List entity types and their attributes based on the data requirements

❑ Entity Type 2: ACTOR



- Requirements:
  - Actors are identified by (<u>name and date of birth</u>) and appear in one or more movies
  - Each actor has a role in the movie and has zero or more quotable quotes to speak
- Attributes:
  - *<u>Akey</u>(Name, Date_of_birth), {Acts_on(Movie, Role, {Quotes})}*



Exercise 1: ER Schema Diagram (Cont.)

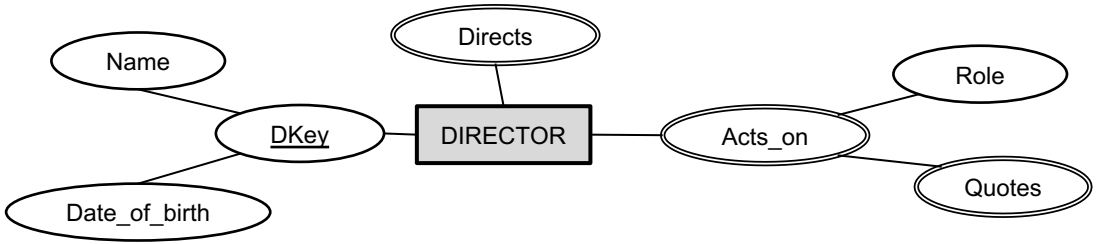
❑ Step 1: Describing the Entity Types

- List entity types and their attributes based on the data requirements

❑ Entity Type 3: DIRECTOR

- Requirements:
  - Directors are identified by (name and date of birth) and direct one or more movies
  - It is possible for a director to act in a movie (including one that she/he may also direct)
- Attributes
  - *Dkey(Name, Date_of_birth), {Directs}, {Acts_on(Movie, Role, {Quotes})}*





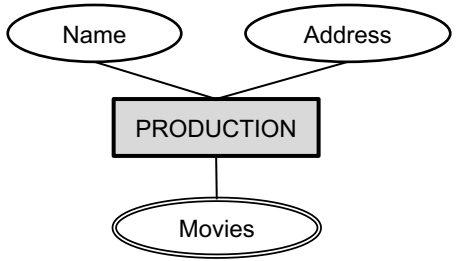
Exercise 1: ER Schema Diagram (Cont.)

Step 1: Describing the Entity Types

- List entity types and their attributes based on the data requirements

Entity Type 4: PRODUCTION

- Requirements:
  - Production companies are identified by (name) and each has an address
  - A production company produces one or more movies
- Attributes:
  - *Name, Address, {Movies}*

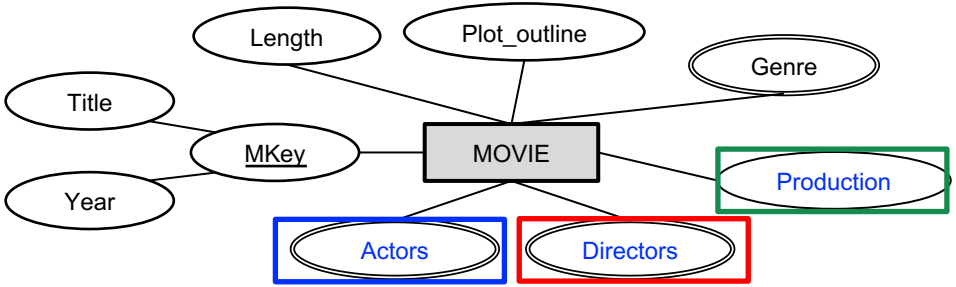


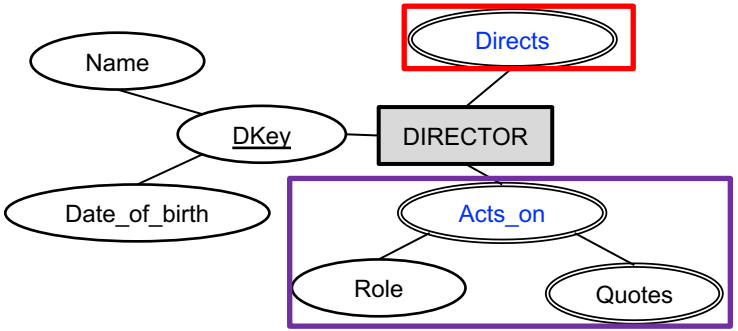
Exercise 1: ER Schema Diagram (Cont.)

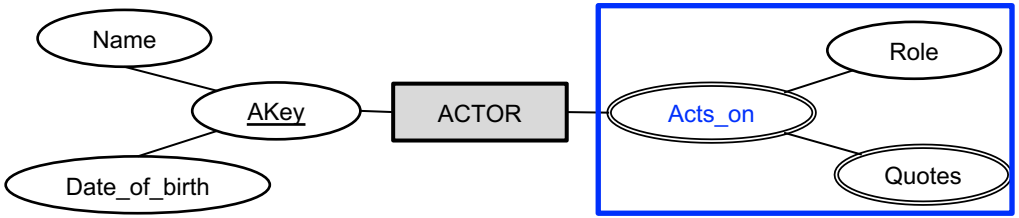
❑ Step 2: Describing the Relationship Types

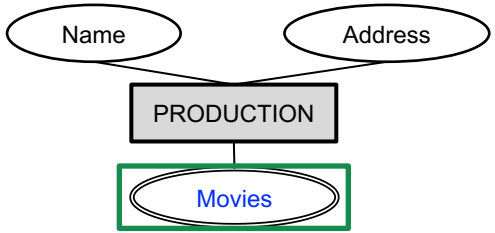
- Specify relationship types and their constraints

❑ How Many Relationship Types Exist?









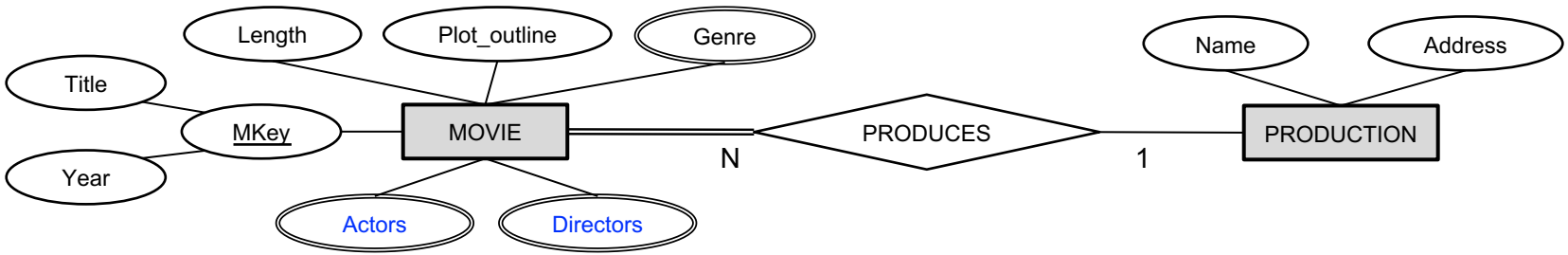
Exercise 1: ER Schema Diagram (Cont.)

Step 2: Describing the Relationship Types

- Specify relationship types and their constraints

Relationship Type 1: PRODUCES

- Relating **PRODUCTION** and **MOVIE** entity types
  - A single production company can produce many movies, but a movie has to be produced by a single production company
    - i.e., Cardinality ratio of this relationship type: (1:N)
  - The MOVIE entity type has a total participation constraint



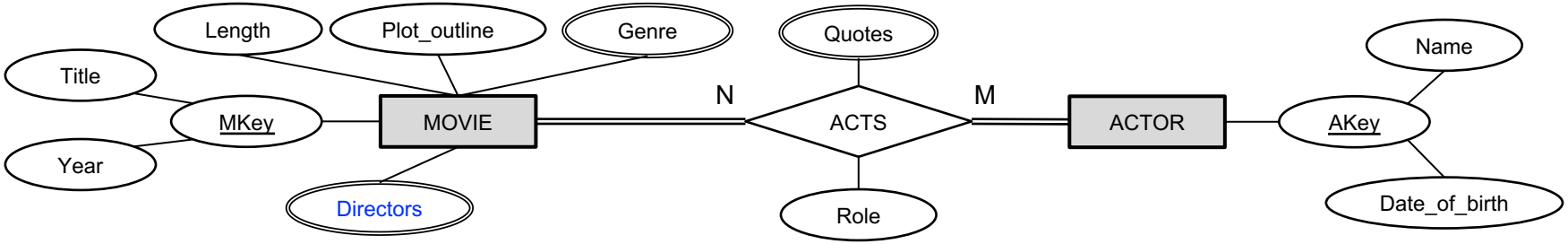
Exercise 1: ER Schema Diagram (Cont.)

❑ Step 2: Describing the Relationship Types

- Specify relationship types and their constraints

❑ Relationship Type 2: ACTS

- Relating **ACTOR** and **MOVIE** entity types
  - ❑ An actor can act in many movies, and a movie can have many actors
    - i.e., Cardinality ratio of this relationship type: (M:N)
  - ❑ This relationship type has two relationship attributes *Role* and *Quotes* (the multivalued attribute)
    - The *Role* attribute names a role that the actor plays in the movie, and the *Quotes* are quotes spoken by an actor
  - ❑ The both entity types have a total participation constraint



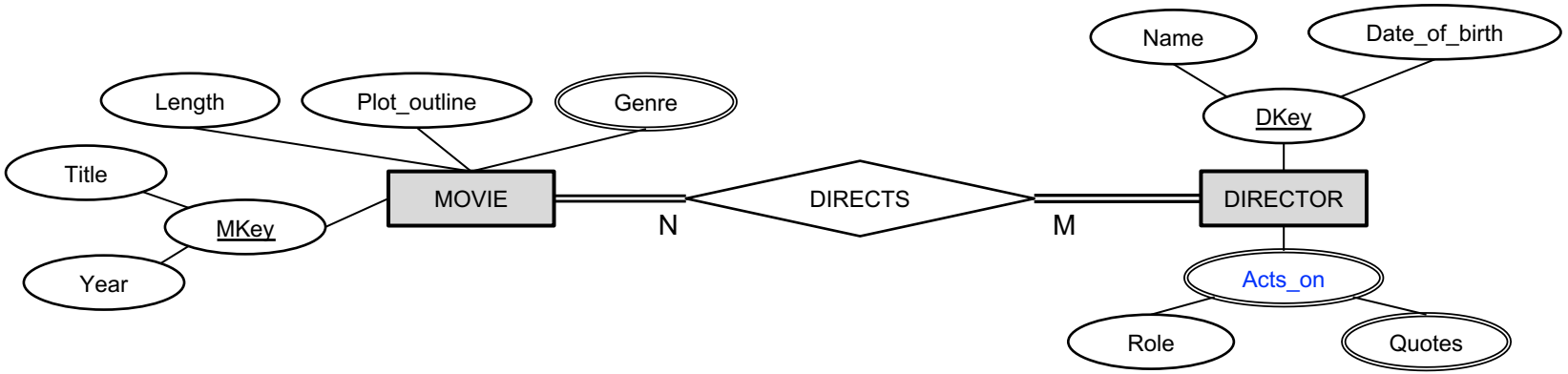
Exercise 1: ER Schema Diagram (Cont.)

Step 2: Describing the Relationship Types

- Specify relationship types and their constraints

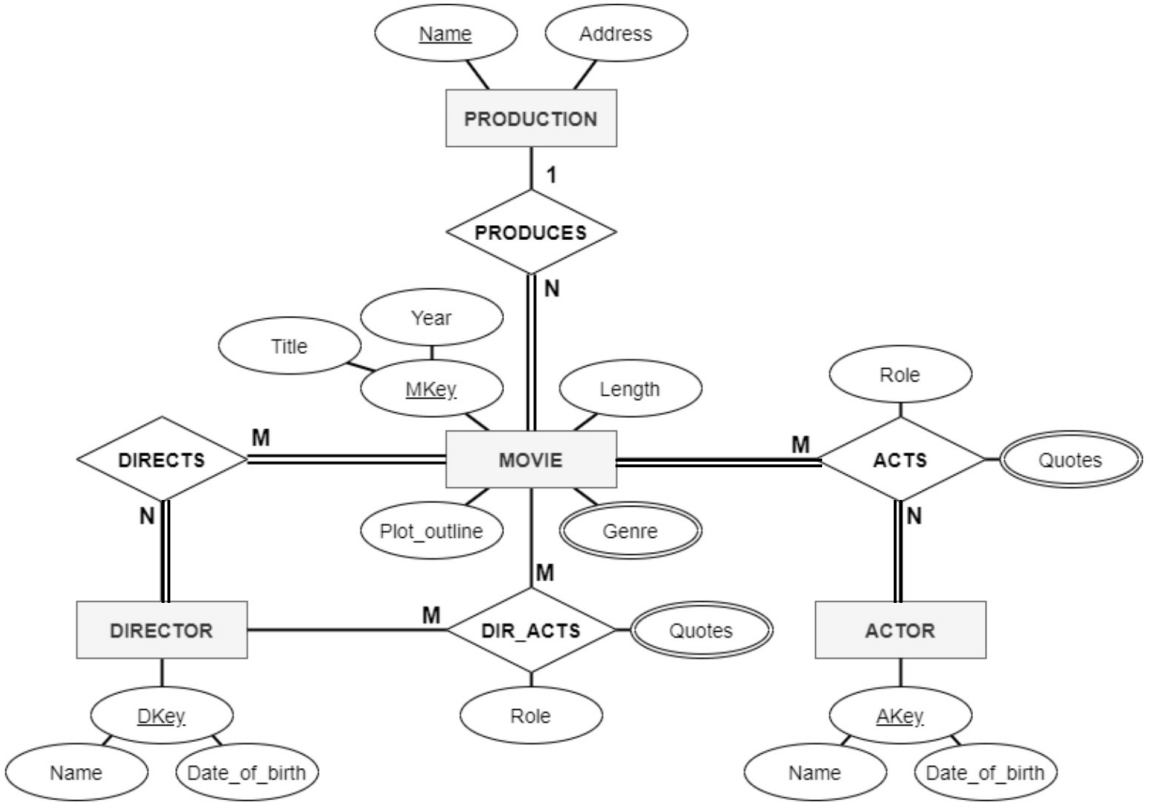
Relationship Type 3: DIRECTS

- Relating **DIRECTOR** and **MOVIE** entity types
  - A director can direct many movies, and a movie is directed by at least one director
    - i.e., Cardinality ratio of this relationship type: (M:N)
  - The MOVIE entity type has a total participation constraint



Exercise 1: ER Schema Diagram (Cont.)

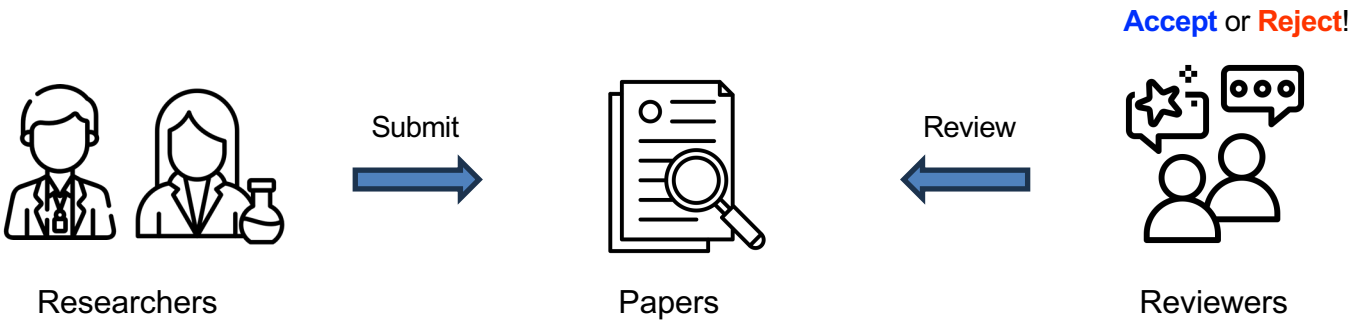
❑ Step 3: Drawing the ER Diagram



Exercise 2: ER Schema Diagram

❑ Exercise 2: Design ER Diagram for CONF_REVIEW Database

- Consider a **CONFERENCE REVIEW** database in which researchers submit their research papers for publication
- Reviews by reviewers are recorded for use in the paper selection process
- The database system caters primarily to reviewers who record answers to evaluation questions for each paper they review and make recommendations regarding whether to accept or reject the paper.



Exercise 2: ER Schema Diagram (Cont.)

Designing ER Diagram for CONF_REVIEW Database

- The data requirements are summarized as follows:
  - AUTHOR
    - Authors of papers are uniquely identified by e-mail id and have the first and last names
    - Authors can submit one or more papers
    - An author should submit a paper to be recorded
  - PAPER
    - Each paper is assigned a unique id by the system and is described by a title, abstract, and the file name
    - Each can have multiple authors, but one of the authors should be designated as the contact author
    - A paper is assigned at least two and at most four reviewers
  - REVIEWER
    - Reviewers of papers are uniquely identified by e-mail id
    - Every reviewer has to review at least one paper to be recorded and can review many papers
    - Each reviewer has the first and last name, phone number, affiliation, and topics of interest
    - A reviewer rates each paper assigned to her/him on a scale of 1 to 10 in four evaluation categories: (1) technical merit, (2) readability, (3) originality, and (4) relevance to the conference.
    - Each reviewer provides an overall recommendation to each paper assigned to her/him
    - Each review contains two types of written comments: (1) one to be seen by the review committee only and (2) the other as feedback to author(s)

Exercise 2: ER Schema Diagram (Cont.)

❑ Step 1: Describing the Entity Types

- List entity types and their attributes based on the data requirements

❑ Step 2: Describing the Relationship Types

- List and describe relationship types and their cardinality ratios

❑ Step 3: Drawing the ER Diagram

Course Schedule (Tentative)

Week		Topic	Week		Topic
1	~~2026.09.01~~	~~Course Introduction~~	9	2026.10.27	AI for Databases: NL2SQL
	~~2026.09.04~~	~~No Class~~		2026.10.30	Hands-on Practice
2	~~2026.09.08~~	~~ER Model (Video Lecture)~~	10	2026.11.03	AI for Databases: NL2SQL
	2026.09.11	**Lab: ER Modeling**		2026.11.06	Hands-on Practice
3	2026.09.15	Relational Database (Video Lecture)	11	2026.11.10	Disk Storage and File Structure
	2026.09.18	Lab: SQL Setup		2026.11.13	Parallelizing Disk Access
4	2026.09.22	Basic SQL (Video Lecture)	12	2026.11.17	Indexing for Physical Database Design (Assignment #2)
	2026.09.25	Lab: SQL DDL (No Class, 추석연휴)		2026.11.20	Advanced Indexing Techniques
5	2026.09.29	Advanced SQL (Assignment #1)	13	2026.11.24	Query Processing
	2026.10.02	Lab: SQL Retrieval		2026.11.27	Query Optimization
6	2026.10.06	Relational Algebra	14	2026.12.01	Transaction Processing
	2026.10.09	Lab: SQL Retrieval (No Class, 한글날)		2026.12.04	Concurrency Control and Recovery
7	2026.10.13	Functional Dependencies and Normalization	15	2026.12.08	Advanced Topic: TBD
	2026.10.16	Office Hours (Q&A)		2026.12.11	No class (Q&A)
8	2026.10.23	**Midterm Exam**	16	2026.12.18	**Final Exam**

Q&A:
ER Model (Hands-on Practices)